

FIN 295
Formula Sheet – Final Exam

Chapter 3

Net Working Capital = Current Assets – Current Liabilities

Net Operating Working Capital = Operating Current Assets – Operating Current Liabilities

Net Operating Working Capital = (Current Assets - Excess Cash) – (Current Liabilities – Notes Payable)

FCF = [EBIT(1 – T) + Depreciation and amortization] – [Capital expenditures + Δ NOWC]

Market value added (MVA) = ($P_0 \times$ Number of shares) – Book value.

Economic value added (EVA) = EBIT(1 – T) – (Total invested capital $\times$ Cost of capital)

Chapter 4

Current ratio = Current assets/Current liabilities

Quick ratio = (Current assets – Inventories) / Current liabilities

Cash ratio = Cash/Current liabilities

Inv. turnover = COGS/Inventories

DSO = Receivables/Avg. sales per day

TA turnover = Sales/Total assets

FA turnover = Sales/Net fixed assets

Debt-to-capital ratio = Total debt / Total invested capital

Operating Margin = EBIT/Sales

Profit Margin = Net Income/Sales

Basic Earning Power = EBIT/Total Assets

ROA = Net Income/Total Assets

ROA = Net Income/Total Equity

P/E = Price/Earnings per share

M/B = Market price/Book value per share

Enterprise Value = MVE + MVD + MVClaims – (Cash and Equivalents)

ROE = (NI/Sales) $\times$ (Sales/TA) $\times$ (TA/Equity)

Chapter 5

$PV = FV_N / (1 + I)^N$

$FV_N = PV \times (1 + I)^N$

$EFF\% = (1 + I_{NOM}/M)^M - 1$

Chapter 6

$r = r^* + IP + DRP + LP + MRP$

$IP_N = \frac{\sum_{t=1}^N INFL_t}{N}$

$MRP_t = 0.1\% (t - 1)$

Corporate bond yield spread = Corporate bond yield – Treasury bond yield
= DRP + LP

Chapter 7

$$\text{Bond Price} = \frac{\text{Coupon}}{(1+r)^1} + \frac{\text{Coupon}}{(1+r)^2} + \dots + \frac{\text{Coupon} + \text{Par Value}}{(1+r)^N}$$

$$\text{Current yield} = \frac{\text{Annual coupon pmt}}{\text{Current price}}$$

$$\text{Capital gains yield} = \frac{\text{Change in price}}{\text{Beginning price}}$$

$$\text{Exp total return} = \text{YTM} = \text{Exp Curr yld} + \text{Exp cap gains yld}$$

Chapter 8

$\hat{r}$ = Expected rate of return

$$\hat{r} = \sum_{i=1}^N P_i r_i$$

σ = Standard deviation

$$\sigma = \sqrt{\text{Variance}} = \sqrt{\sigma^2}$$

$$\sigma = \sqrt{\sum_{i=1}^N (r_i - \hat{r})^2 P_i}$$

$$\text{CV} = \frac{\text{Standard deviation}}{\text{Expected return}} = \frac{\sigma}{\hat{r}}$$

$\hat{r}_p$ is a weighted average:

$$\hat{r}_p = \sum_{i=1}^N w_i \hat{r}_i$$

$$r_i = r_{RF} + (r_M - r_{RF}) b_i$$

Chapter 9

$$\hat{P}_0 = \frac{D_1}{(1+r_s)^1} + \frac{D_2}{(1+r_s)^2} + \frac{D_3}{(1+r_s)^3} + \dots + \frac{D_\infty}{(1+r_s)^\infty}$$

$$\hat{P}_0 = \frac{D_0(1+g)}{r_s - g} = \frac{D_1}{r_s - g}$$

$$\text{Dividend yield} = D_1/P_0$$

$$\text{Capital gains yield} = (P_1 - P_0)/P_0$$

$$\text{Total return } (r_s) = \text{Dividend yield} + \text{Capital gains yield}$$

$$\text{Preferred stock: } V_p = D/r_p$$

Chapter 10

$$\text{WACC} = w_d r_d (1 - T) + w_p r_p + w_c r_s$$

$$r_p = D_p / P_p$$

$$\text{CAPM: } r_s = r_{\text{RF}} + (r_M - r_{\text{RF}})b$$

$$\text{DCF: } r_s = (D_1 / P_0) + g$$

$$\text{Bond-Yield-Plus-Risk-Premium: } r_s = r_d + \text{RP}$$

$$r_e = \frac{D_0(1+g)}{P_0(1-F)} + g$$

Chapter 11

$$\text{NPV} = \sum_{t=0}^N \frac{CF_t}{(1+r)^t}$$

IRR:

$$0 = \sum_{t=0}^N \frac{CF_t}{(1+\text{IRR})^t}$$